

ANDREW ROOT

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PROFESSIONAL EXPERIENCE

Founding Engineer — *Daentra (Startup)* — San Francisco, CA 08/25 - Present

- Built and deployed an end-to-end vision/audio system including sensors, edge devices running real-time inference, and cloud infrastructure; now live in dental operatory rooms across 2 offices.
- Research and prototype depth and pose estimation methods for key-object tracking in video, evaluating state-of-the-art models and implementing custom solutions.
- Designed a novel frame-timing synchronization algorithm to align a machine vision camera with a high-intensity flickering light of varying frequency and color.

Research Lab Assistant — *Oakland University* — Oakland County, MI 05/24 - 05/25

- Developed a real-time machine learning pipeline to detect foreign objects in video streams for an automated security vehicle system (in collaboration with Dataspeed Inc.).

Undergraduate Researcher — *University of Wisconsin Eau Claire* — Eau Claire, WI 09/23 - 12/24

- Conducted cyberbullying-detection research investigating sources of bias in ML datasets through cross-dataset evaluation on the university HPC cluster; mentored junior researchers.

Software Engineer Intern — *Midwest Manufacturing (Menards Inc.)* — Eau Claire, WI 05/23 - 05/24

- Developed and deployed a web application that streamlined advanced database operations for technical and non-technical teams.
- Refactored manufacturing services integrating PLC systems and message brokers, and automated documentation updates via web scraping and syntax parsers.

EDUCATION

B.S. Computer Science & Applied Math (Double) — UW Eau Claire (3.98/4.0) 09/22 - 12/24

B.S. Computer Science — University of Minnesota - Twin Cities 09/18 - 12/19

TECHNICAL SKILLS

Langs & Libraries	Python, C++, TypeScript, PyTorch, TensorFlow, OpenCV, Open3D, AWS CDK
Hardware	Raspberry Pi Pico 2W, Orange Pi 5B, Machine Vision Cameras (e.g. See3CAM.24CUG)
Other	AWS (e.g. SageMaker), Git/Github, Linux, CAD

HONORS AND AWARDS

Graduated with Honors in Computer Science — *UW Eau Claire* 12/24

Fred F. and Lorain E. Winrich Scholarship — *UW Eau Claire* 09/23 - 12/24

EXTRACURRICULAR & PROJECTS

President of the Students' Association for Computing Machinery — Designed and led interactive and educational events for my schools chapter of SACM. Event topics included Git/Github, Python, and Neural Networks. Led meetings and delegated assignments to board members.

Instructor for Python & Machine Learning Workshop — Designed and ran a workshop for over 20 UWEC students, on Python, statistics, data analytics, and machine learning.

Voice-Assistant Task Manager 🗣️ — A full-stack, real-time voice-assistant task manager that allows users to create, edit, and manage tasks entirely through natural language without traditional UI controls. The system streams audio to a Python agent that interprets commands with speech-to-text and LLMs, executes database operations, and synchronizes updates to a Next.js frontend in real time.